



Transcend

<https://www.transcendh2o.com>

Transcend: A Breakthrough Technology

Generative Design SaaS for conceptual and preliminary engineering



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THE PROBLEM

As the human race faces the unprecedented challenge of climate change, we must rapidly develop new **sustainable infrastructure** to repair our broken planet and sustain our quality of life without further destroying the environment. Many of the required technologies exist today, often w/ superior economics to the status quo. **So why aren't we using them?**

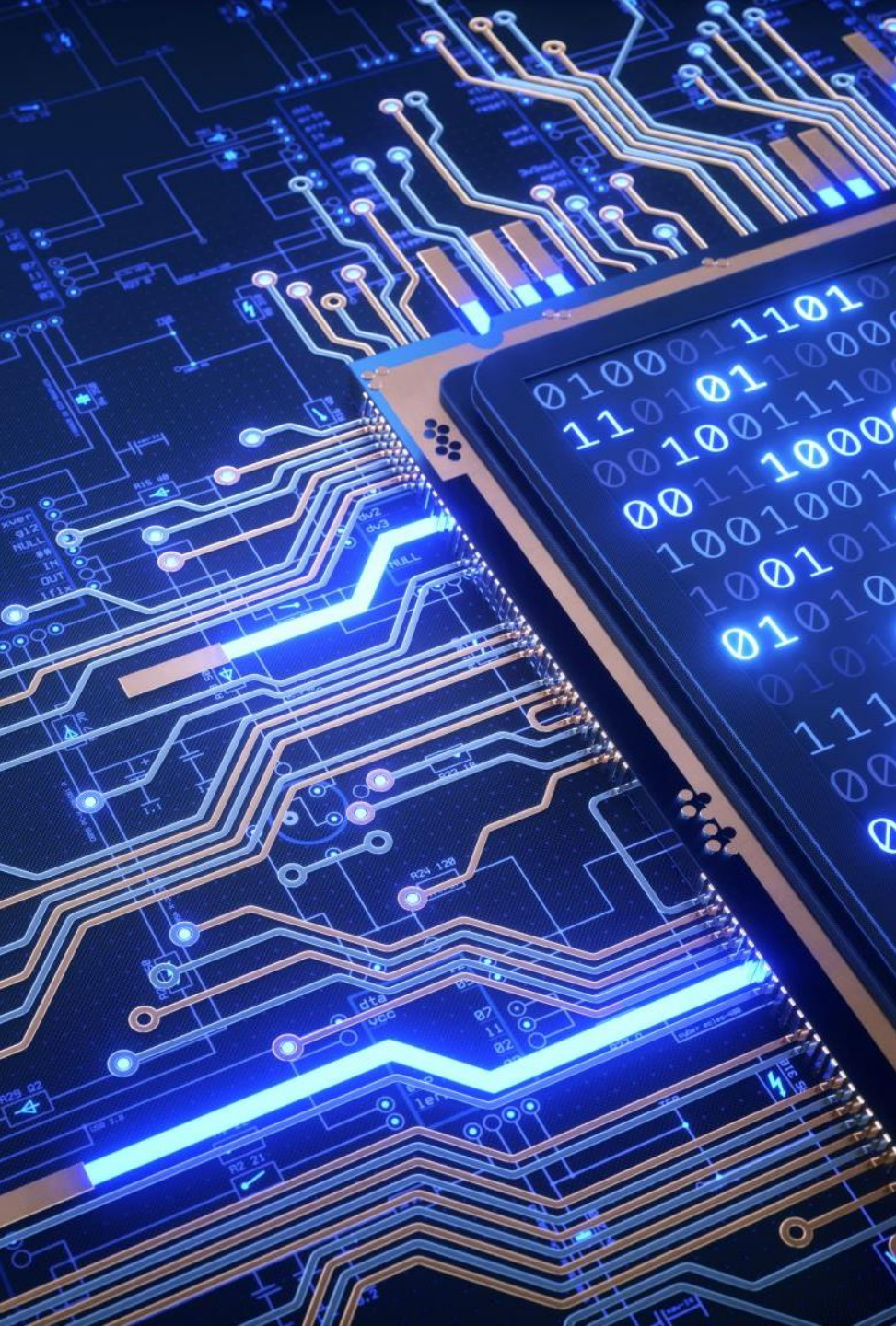
One major reason is the way the industry approaches infrastructure design. Today's **manual approach to conceptual design** is both time and labor intensive, resulting in a significant financial burden for industry players. The result is that asset owners struggle to get an **accurate budget** and are **limited in their ability to run scenarios, innovate**, and integrate already existing solutions into projects that can help solve our climate crisis.

THE SOLUTION

Transcend's mission is to transform the way utilities/ asset owners, engineering firms, and technology OEMs **assess and design critical infrastructure and vertical assets**. Transcend was started to help change the culture of these firms by giving engineers **the time and resources** to evaluate innovative technologies and sustainable design approaches, rather than the standard, sub-optimal solutions of years past.

We are a team of successful entrepreneurs and technologists that has developed revolutionary engineering automation software. By **driving down the costs and resources required in the conceptual design stages of a project**, and improving the velocity and level of detail in client responses, we will help bring the infrastructure industry into the 21st Century

**THE FOUNDATION OF A PROFITABLE INFRASTRUCTURE
PROJECT IS FAST AND ACCURATE PRELIMINARY
ENGINEERING & ESTIMATION**



Transforming the Infrastructure Industry

Transcend's cloud-based SaaS platform, the Transcend Design Generator (TDG), **fully automates the preliminary engineering process** for critical infrastructure projects and other kinds of vertical assets, including water and wastewater treatment, electrical substations, and power generation.

TDG produces professional grade preliminary engineering documents – ~30% of a complete design - with no human intervention. The software uses bleeding edge AI/ML and other algorithms, **trained and coded by professional engineers**, to complete all required calculations and instruct EXISTING industry-standard engineering software tools (simulators, CAD, BIM, etc) to generate optimized designs.

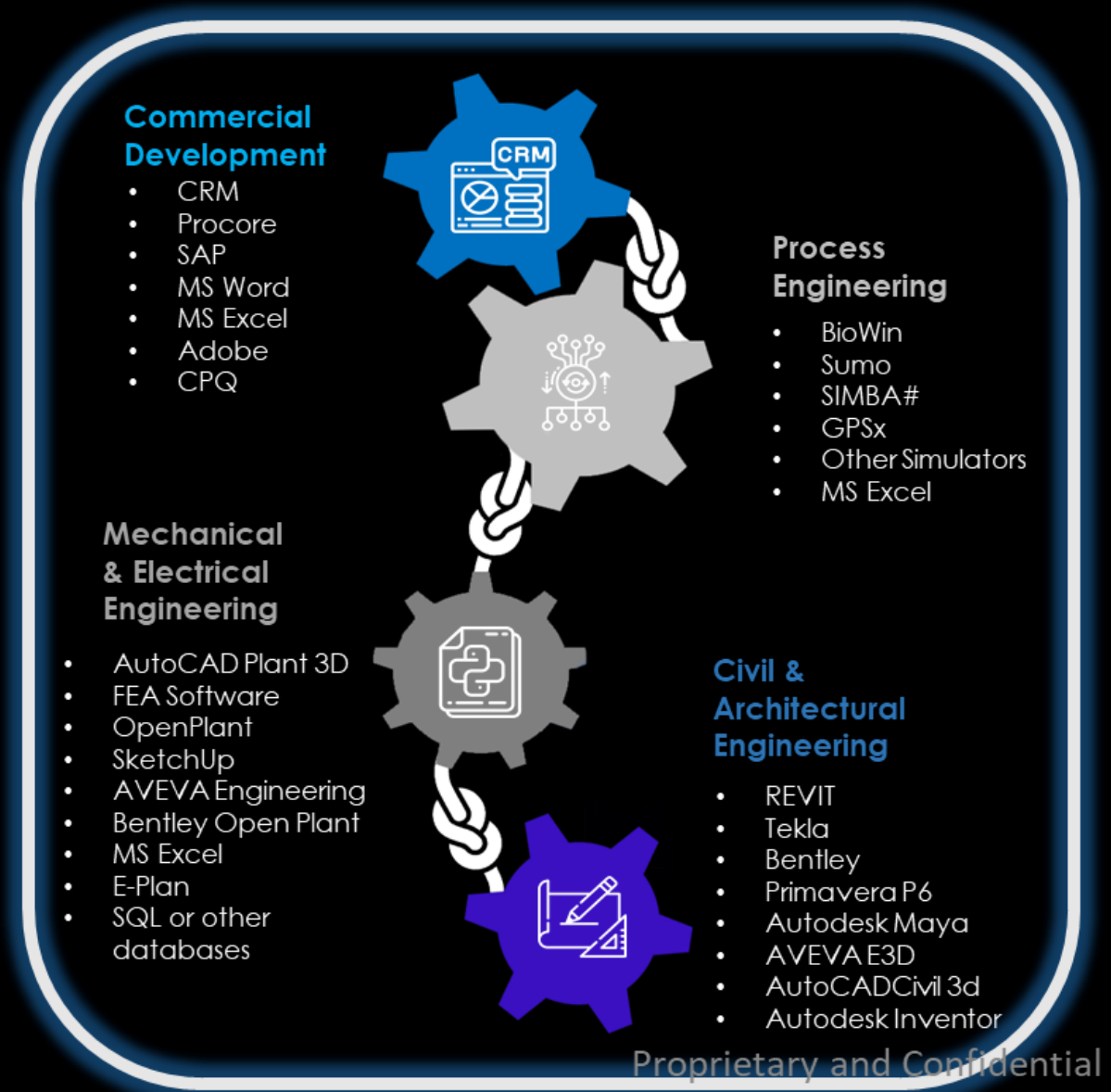
These 30% designs are generated in a **fraction of the time** and w/ **90% less resources** compared to traditional processes. This dramatically reduces fixed overhead cost, speeds up project cycles, and frees up resources to bid on more projects with more detail, and consider more options prior to bidding and/or fixing a project budget.

Platform Connecting Design Decisions

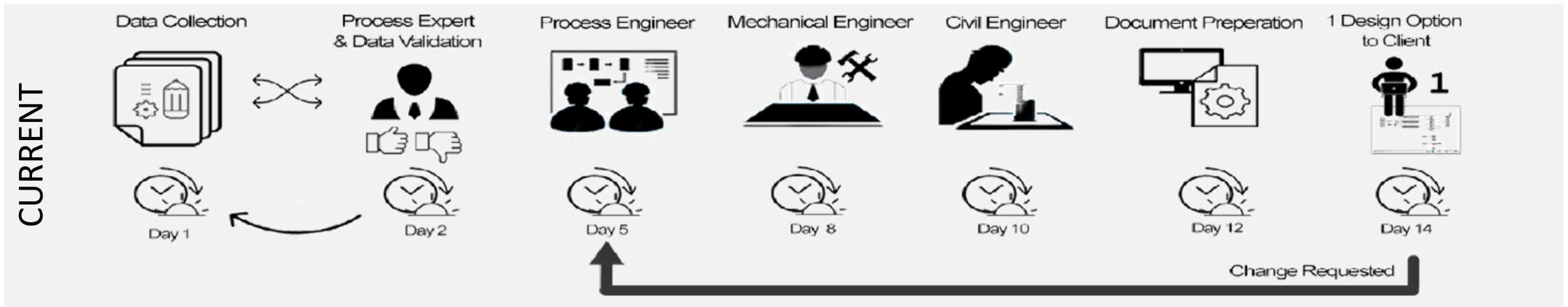
Augment and Integrate Existing Software, Tools, and Processes

- Acts as a thread tying all water, power, and other critical infrastructure engineering disciplines together in a single platform for Conceptual and Preliminary design of vertical assets
- Augments existing software and people to avoid starting with the last project or template, and instead generate a custom design automatically each time:
 - **30% designs are done in hours not weeks**
 - **Run multiple design options** and select the most CAPEX/OPEX/Carbon efficient one
 - **Use automatically generated design outputs (BIM, Smart P&ID) as starting point for detailed design**

Completely Automated Preliminary Design Process with Transcend

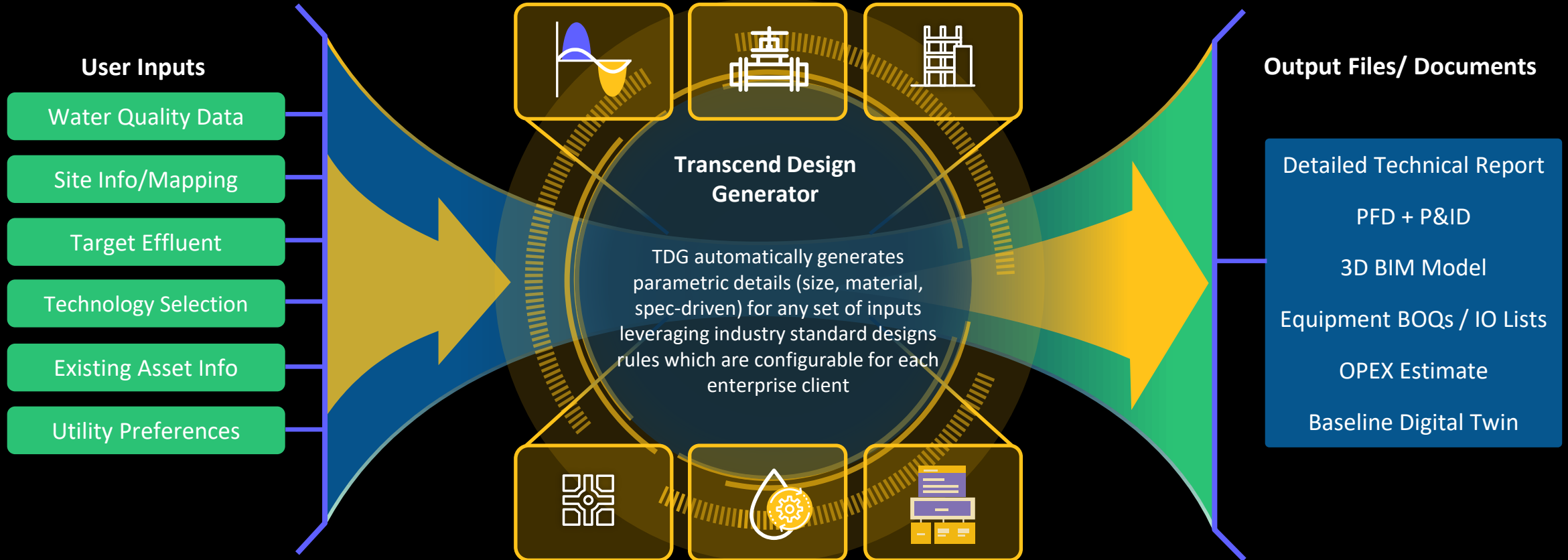


Preliminary Design Phase Transformation



-  Substantial time and money savings
-  Better management of team capabilities
-  More options drive better outcomes
-  Consistent quality & compliance w/ standards

Unlocking the Door to Generative Design for Vertical Assets



Transcend History

Created to solve a real business need prevalent across the industry



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Ideation to Product Launch

- Software POC started
- Completed V 1.0 to automate design of Organica- powered WWTPs
- Integrated engineering team into software development and QA/QC

Product-Market Fit Validation

- Utilization by Organica commercial and engineering teams (2014+)
- Addition of other wastewater technologies/unit processes (2015+)
- ~2000+ designs run

Commercial Validation

- Product registration opened to 3rd party ESPs and Organica License Partners
- Launched V 2.0 and migrated to AWS
- 4000+ designs run

Spin Out and Seed Financing

- TRANSCEND Brand Release
- Hired first dedicated employee
- Spin Out from Organica (Dec 2019) and initial seed financing

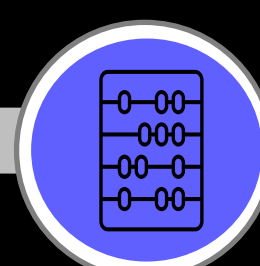
2011-2013



2014-2016



2017-2018



2019-2020





BLACK & VEATCH



Utilities

EPCs

OEMs

Top Tier Customers and Industry Recognition



2021 Hottest Innovation to Watch
2022 Breakthrough Technology
Company of the Year Finalist



2021: Precon Planning



2021 Innovation in Optioneering



Cohort 9 Graduate



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Existing Enterprise Clients: TDG Allows Cost-Effective and Scalable Growth of Engineered Product Sales



Kubota Membrane USA Corporation is a Membrane Bioreactor (MBR) system supplier utilizing the Kubota Submerged Membrane Unit® (SMU). The Kubota SMU was developed to treat municipal, commercial and industrial wastewater simply and successfully.

Background: Kubota sells indirectly through regional sales representatives that are supported by Kubota Engineering and commercial staff. Kubota is in a growth mode and looking to bid on many projects.

Needs: Kubota needs a software that will allow it to bid more projects without having to hire more people or slow the sales process down.

Type: Technology OEM

Status: Contracted October 2020

“Our commercial team will be able to bid more and not need as much engineering time” – Lead Application Engineer



Dupont Water is unmatched in global reach, industry experience, and expertise. Products include reverse osmosis, nanofiltration, desalination, MABR, and ultrafiltration membranes, ion exchange, and electrodeionization.

Background: Dupont continues to acquire additional technology companies around the world. Over the last 12 months Dupont has done four deals that have brought 4 new unique offerings to their portfolio.

Needs: Dupont wants to increase market share in the municipal wastewater industry, without increasing their cost to serve. This industry needs a higher touch and has longer sales cycle than industrial. Design Generator will enable them to present their portfolio of technologies within a complete facility.

Type: Technology OEM

Status: Contracted December 2020

“We will be able to enter the municipal market and provide more than just the technology.... Value added engineering” – Regional Commercial Director



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Existing Enterprise Clients: TDG Serves as Competitive Differentiator for Large Engineering Firms



Black & Veatch is a **global engineering, procurement, construction and consulting company**. Black & Veatch is engaged in every facet of the water cycle. Helping clients efficiently and effectively collect, store, move and treat water and wastewater; plan and manage watersheds; harness the energy embedded in the treatment and movement of water; and enhance sustainability.

Headquarters: Overland Park, KS

Revenue: 3.7 billion USD

Number of employees: 10,400

Status: Lighthouse client; Phase 1 completed and Phase 2 Contract started

Background: B&V currently completes 10,000+ wastewater treatment facility designs every year. Each design takes “~1500 engineering hours to complete,” according to B&V Chief Engineer Michael Etheridge.

Needs: B&V is looking for a solution to help them become more competitive in their ability to engineer, procure, and construct WWTP’s. The solution needed to automate their engineering team’s non-value added (and non-billable) activities and allow them to offer more, higher quality alternatives to their clients in less time.

“We have tried to do this for 15 years with no success” – Global Chief Engineer



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Enterprise Pipeline: TDG Allows Asset Owners to Reduce Engineering Spend w/o Reducing Bid Volume



BRK Ambiental is the largest private sanitation company in the Brazil owning/operating water and sewage systems in 12 states in Brazil, providing a service that benefits the lives of 15 million people.

Background: BRK is growing via winning new water concessions and investing in new and upgraded wastewater treatment facilities with long term contracts.

Needs: BRK needs to be able to be fast and flexible to evaluate bidding opportunities and win new deals. Today they outsource the engineering design and with TDG they can dramatically lower those costs by bringing that work in house, also improving speed, flexibility, and ROI via more optionality assessments.

Type: EPC/Asset Owner

Status: Contracted in March 2021

"Transcend helped us reduce our third-party consulting spend by over half and enabled us to win a \$2B concession contract."

- Engineering and Technology Director



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